The empirical recurrence for OEIS A300877, recovered from its tail

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Abstract

OEIS A300877 records "Empirical recurrence of order 97" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 111$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 99$. Its last coefficient is $c_{97} = 1 \neq 0$, so no further tail satisfies anything shorter; any order-97 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A300877 is "Number of nX4 0..1 arrays with every element equal to 0, 1, 2, 4 or 5 horizontally, vertically or antidiagonally adjacent elements, with upper left element zero.". It has offset 1 and begins

$$8, 50, 148, 589, 2264, 8739, 34693, 138559, \ \ldots$$

The entry states:

> Empirical recurrence of order 97 (see link above)

As of the "Last modified" line on the live entry (Mar 14 2018 07:18:00, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 111$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 111$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 97: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 97.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 222$ exact terms from $n = 2$ on, it returns order 97 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{97} c_i\, a(n-i),$$

c_1	12	c_{26}	-426610	c_{51}	88152993	c_{76}	1005151
c_2	-58	c_{27}	-298304	c_{52}	207492187	c_{77}	373142
c_3	157	c_{28}	1074564	c_{53}	19570833	c_{78}	395273
c_4	-325	c_{29}	1442413	c_{54}	-211044324	c_{79}	176950
c_5	709	c_{30}	-3954467	c_{55}	-97007371	c_{80}	-46963
c_6	-1400	c_{31}	-7657443	c_{56}	40026373	c_{81}	-82255
c_7	1953	c_{32}	-22547	c_{57}	-92647890	c_{82}	-129576
c_8	-2795	c_{33}	8107851	c_{58}	-127136404	c_{83}	-68649
c_9	5735	c_{34}	1889467	c_{59}	53382635	c_{84}	-20332
c_{10}	-8109	c_{35}	-8744053	c_{60}	105844994	c_{85}	12408
c_{11}	6884	c_{36}	-2269522	c_{61}	20941180	c_{86}	14198
c_{12}	-11853	c_{37}	23780822	c_{62}	-4823072	c_{87}	4594
c_{13}	22005	c_{38}	21691527	c_{63}	7037006	c_{88}	-555
c_{14}	-13070	c_{39}	-7273291	c_{64}	5279533	c_{89}	-1626
c_{15}	11872	c_{40}	12537142	c_{65}	300139	c_{90}	-310
c_{16}	-36694	c_{41}	33678853	c_{66}	-3131195	c_{91}	220
c_{17}	14107	c_{42}	-19810015	c_{67}	-7939115	c_{92}	174
c_{18}	-17561	c_{43}	-87851704	c_{68}	-8332851	c_{93}	28
c_{19}	63219	c_{44}	-53213741	c_{69}	-4547640	c_{94}	-24
c_{20}	9931	c_{45}	31970693	c_{70}	2509500	c_{95}	-8
c_{21}	-6396	c_{46}	-48873423	c_{71}	3953071	c_{96}	-1
c_{22}	-279095	c_{47}	-130944641	c_{72}	-2183166	c_{97}	1
c_{23}	55550	c_{48}	54032302	c_{73}	-1827474		
c_{24}	389905	c_{49}	153850094	c_{74}	24998		
c_{25}	588485	c_{50}	27803730	c_{75}	686456		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 99$, and it is the recurrence OEIS A300877 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{97} - c_1 x^{97-1} - \cdots - c_{97}$. Its constant term is $-c_{97} = -1$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 97 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 97, so $\deg q = 97$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 97, so $q = q_{\min}$. □

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 22 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 97, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 1.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-97 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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